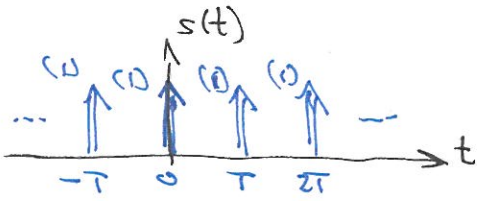
# Sampling

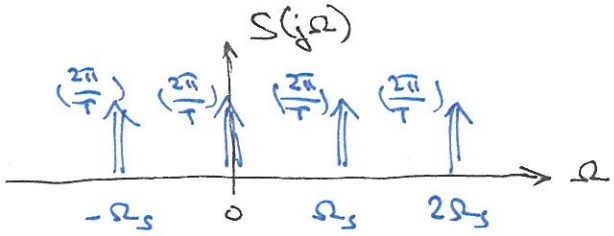
$x_s(t) = x_c(t) \cdot s(t)$ sampling of CT waveform $x_c(t)$

$\Omega_s = \frac{2\pi}{T} = 2\pi f_s$

$\omega = \frac{\Omega}{f_s} = \Omega T$

$$s(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT) \iff S(j\Omega) = \frac{2\pi}{T} \sum_{k=-\infty}^{\infty} \delta(\Omega - k\Omega_s)$$




$$x_s(t) = x_c(t) \sum_n \delta(t - nT)$$
$$= \sum_n x_c(nT) \cdot \delta(t - nT)$$
$$= \sum_n x[n]\, \delta(t - nT)$$

$$X_s(j\Omega) = \frac{1}{2\pi} X_c(j\Omega) * S(j\Omega)$$
$$= \frac{1}{T} \sum_k X_c(j(\Omega - k\Omega_s))$$

$\Updownarrow$

$$X_s(j\Omega) = \mathcal{F}\left\{\sum_n x[n]\, \delta(t - nT)\right\}$$
$$= \sum_n x[n] \cdot \mathcal{F}\{\delta(t - nT)\}$$
$$= \sum_n x[n]\, e^{-j\Omega T n} = X(e^{j\omega})\Big|_{\omega = \Omega T} = X(e^{j\Omega T})$$

$$\Rightarrow X_s(j\Omega) = \frac{1}{T} \sum_k X_c(j(\Omega - k\Omega_s)) = X(e^{j\Omega T}) = X(e^{j\omega})\Big|_{\omega = \Omega T}$$

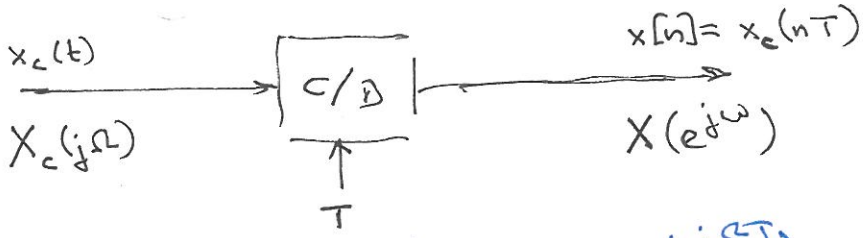
$\Rightarrow$ or

$$\boxed{X(e^{j\Omega T}) = \frac{1}{T} \sum_k X_c(j(\Omega - k\Omega_s))}$$
$$\boxed{X(e^{j\omega}) = \frac{1}{T} \sum_k X_c\left(j\left(\frac{\omega}{T} - \frac{2\pi k}{T}\right)\right)}$$

$\Omega = \frac{\omega}{T}$

$$s(t) = \sum_n \delta(t - nT) \underset{\text{IFS}}{=} \sum_k S[k]\, e^{j\pi k t/T} \iff S(j\Omega) = \mathcal{F}\left\{\sum_k S[k]\, e^{j\pi k t/T}\right\}$$
$$= \sum_k S[k]\, \mathcal{F}\{e^{j\pi k t/T}\}$$

# C to D Conversion




from $X_s(j\Omega) = X(e^{j\omega})\big|_{\omega=\Omega T} = X(e^{j\Omega T})$

$$X(e^{j\Omega T}) = \frac{1}{T}\sum_k X_c\left(j(\Omega - k\Omega_s)\right)$$

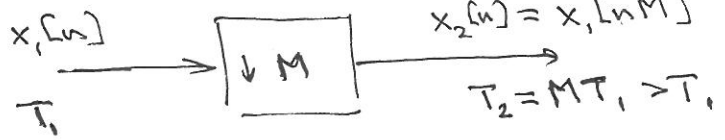
$$X(e^{j\omega}) = \frac{1}{T}\sum_k X_c\left(j\left(\frac{\omega}{T} - \frac{2\pi k}{T}\right)\right)$$

$\Omega_s = \frac{2\pi}{T}$

$\Omega = \frac{\omega}{T}$

Downsampling

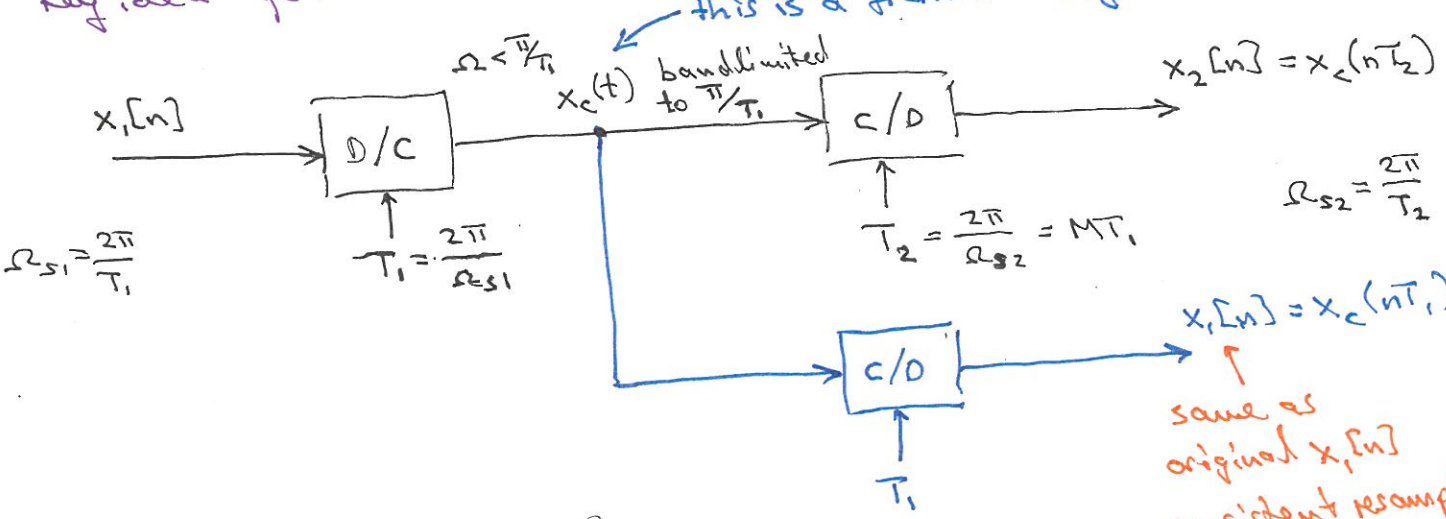
Decimator by M




This can create aliasing

How to analyze?

Key idea for mathematical analysis:




Now

$$X_1(e^{j\omega_1}) = \frac{1}{T_1}\sum_k X_c\left(j\left(\frac{\omega_1}{T_1} - \frac{2\pi k}{T_1}\right)\right) \qquad \omega_1 = \Omega T_1$$

$$X_2(e^{j\omega_2}) = \frac{1}{T_2}\sum_k X_c\left(j\left(\frac{\omega_2}{T_2} - \frac{2\pi k}{T_2}\right)\right) \qquad \omega_2 = \Omega T_2$$

## Downsampling (cont'd.)

$$X_1(e^{j\omega_1}) = \frac{1}{T_1}\sum_k X_c\left(j\left(\frac{\omega_1}{T_1} - \frac{2\pi k}{T_1}\right)\right) \qquad \omega_1 = \Omega T_1$$

$$X_2(e^{j\omega_2}) = \frac{1}{T_2}\sum_k X_c\left(j\left(\frac{\omega_2}{T_2} - \frac{2\pi k}{T_2}\right)\right) \qquad \omega_2 = \Omega T_2$$

$$= \frac{1}{MT_1}\sum_l X_c\left(j\left(\frac{\omega_2}{MT_1} - \frac{2\pi l}{MT_1}\right)\right) \qquad \omega_2 = \Omega M T_1 = M\omega_1$$

Let $l = i + kM \quad i = 0, 1, \ldots, M-1 \quad -\infty < k < \infty$

$$X_2(e^{j\omega_2}) = \frac{1}{M}\sum_{i=0}^{M-1}\frac{1}{T_1}\sum_k X_c\left(j\left(\frac{\omega_2}{MT_1} - \frac{2\pi i}{MT_1} - \frac{2\pi kM}{MT_1}\right)\right)$$

$$= \frac{1}{M}\sum_{i=0}^{M-1}\underbrace{\frac{1}{T_1}\sum_k X_c\left(j\left(\frac{\omega_2 - 2\pi i}{M} - 2\pi k\right)/T_1\right)}_{= X_1\left(e^{j(\omega_2 - 2\pi i)/M}\right)}$$

$$= \frac{1}{M}\sum_{i=0}^{M-1} X_1\left(e^{j(\omega_2 - 2\pi i)/M}\right)$$

# Downsampling: Example

